

Parallel & Perpendicular Lines

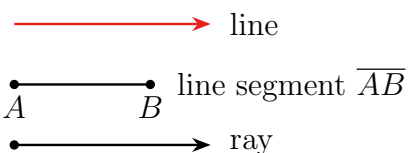
US Common Core – High School Geometry

Domain: Congruence (G-CO.C.9) | Coordinate Geometry (G-GPE.5)

1 Basic Building Blocks

Definitions

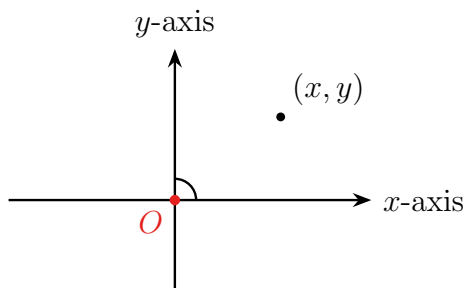
- **Point:** an object that has no length, width, or height – only a location.
- **Line:** a collection of never-ending points extending in both directions.
- **Line segment:** a small portion of a line with two end points.
- **Ray:** has a start point but no end point – it extends infinitely in one direction.



2 Points, Lines and the Coordinate Plane

Coordinate Plane

The **origin** O is the point of intersection between the x -axis and y -axis, which meet at a right angle (90°). Every point in the plane is located by an ordered pair (x, y) .



3 Angles

Definition

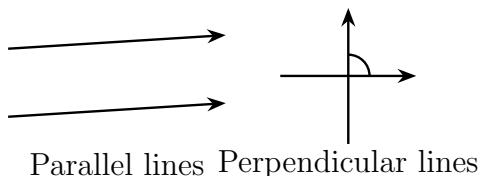
Angles are formed by the intersection of two line segments (or rays), measured as θ (theta), and classified by measure x :

Zero angle	$x = 0^\circ$
Acute angle	$0^\circ < x < 90^\circ$
Right angle	$x = 90^\circ$
Obtuse angle	$90^\circ < x < 180^\circ$
Straight angle	$x = 180^\circ$
Reflex angle	$180^\circ < x < 360^\circ$
Full rotation	$x = 360^\circ$

4 Parallel and Perpendicular Lines

Definitions

Parallel lines are two lines that never meet – the angle between them is 0° . **Perpendicular lines** are two lines that meet at a 90° angle. Two intersecting lines that meet at any other angle are simply **intersecting lines**.



Parallel lines Perpendicular lines

Parallel Postulate

If P is a point not on line ℓ , then exactly one line parallel to ℓ can be drawn through P .

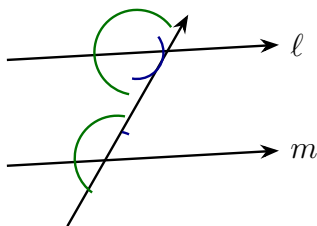
Perpendicular Postulate

If P is a point not on line ℓ , then exactly one line perpendicular to ℓ can be drawn through P .

5 Transversal Lines

Definition

A **transversal** is a line that crosses two (usually parallel) lines. At each intersection it creates an **exterior** region (outside the two lines) and an **interior** region (between the two lines).

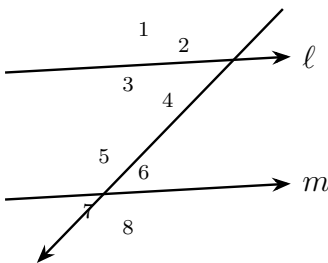


6 Angle Relationships Formed by a Transversal

Named Angle Pairs

Numbering the eight angles 1–8 (top intersection 1, 2, 3, 4; bottom intersection 5, 6, 7, 8, left-to-right, top-to-bottom):

- **Corresponding angles** – same relative position at each intersection: $(\angle 1, \angle 5)$, $(\angle 2, \angle 6)$, $(\angle 3, \angle 7)$, $(\angle 4, \angle 8)$.
- **Alternate interior angles** – interior, on opposite sides of the transversal: $(\angle 3, \angle 6)$, $(\angle 4, \angle 5)$.
- **Alternate exterior angles** – exterior, on opposite sides of the transversal: $(\angle 1, \angle 8)$, $(\angle 2, \angle 7)$.
- **Consecutive (co-interior) angles** – interior, on the same side of the transversal: $(\angle 3, \angle 5)$, $(\angle 4, \angle 6)$.



Why this needs parallel lines

If lines ℓ and m are *not* parallel, these angle pairs are no longer equal (e.g. $\angle 3 \neq \angle 6$). All of the angle relationships above depend on $\ell \parallel m$.

7 Theorem 3.1 – Angle Correspondence Theorem

Theorem

Two angles with the same measure are congruent: if $m\angle ACB = m\angle XYZ$, then $\angle ACB \cong \angle XYZ$.

8 Theorem 3.2 – Alternate Interior Angles Theorem

Theorem

If lines ℓ and m are parallel and cut by a transversal, then alternate interior angles are congruent: $\angle 3 \cong \angle 6$ and $\angle 4 \cong \angle 5$.

9 Theorem 3.3 – Alternate Exterior Angles Theorem

Theorem

If lines ℓ and m are parallel and cut by a transversal, then alternate exterior angles are congruent: $\angle 1 \cong \angle 8$ and $\angle 2 \cong \angle 7$.

10 Theorem 3.4 – Consecutive (Co-Interior) Angles Theorem

Theorem

If lines ℓ and m are parallel and cut by a transversal, then consecutive interior angles are supplementary: $\angle 3 + \angle 5 = 180^\circ$ and $\angle 4 + \angle 6 = 180^\circ$.

Note: The source notes include a further “Theorem 3.9” restating this as consecutive interior angles being equal rather than supplementary. Equal consecutive interior angles is not generally true for parallel lines (only supplementary is guaranteed) – this looks like a slip in the original notes, so it has been folded into Theorem 3.4 here rather than reproduced as a separate, contradictory rule.

11 Converse Theorems 3.5–3.8

Theorem 3.5 (Converse of Angle Correspondence)

If two lines ℓ and m are cut by a transversal and a pair of corresponding angles is congruent, then $\ell \parallel m$.

Theorem 3.6 (Converse of Alternate Interior Angles)

If two lines ℓ and m are cut by a transversal and a pair of alternate interior angles is congruent, then $\ell \parallel m$.

Theorem 3.7 (Converse of Alternate Exterior Angles)

If two lines ℓ and m are cut by a transversal and a pair of alternate exterior angles is congruent, then $\ell \parallel m$.

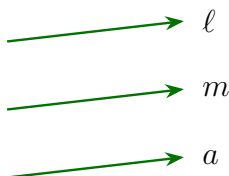
Theorem 3.8 (Converse of Consecutive Interior Angles)

If two lines ℓ and m are cut by a transversal and a pair of consecutive interior angles is supplementary, then $\ell \parallel m$.

12 Theorem 3.10 – Transitivity of Parallel Lines

Theorem

If line ℓ and line m are both parallel to a third line a , then $\ell \parallel m$.



13 Worked Examples: Solving for Unknown Angles

Example 1 (Consecutive interior angles)

Given $\ell \parallel m$: top intersection has angles y° and 75° (a linear pair); bottom intersection has angles x° and 85° (a linear pair).

$$y + 75 = 180 \Rightarrow y = 105^\circ \qquad x + 85 = 180 \Rightarrow x = 95^\circ$$

Example 2 (Alternate/corresponding angles, algebraic)

Given $\ell \parallel m$: one angle is 115° ; its related angle at the other intersection is $(x - 17)^\circ$.

$$x - 17 = 115 \Rightarrow x = 132^\circ$$

Example 3 (Converse of Angle Correspondence)

If a corresponding angle pair measures $(x + 5)^\circ$ and 105° , and these angles are congruent, then the two lines are parallel:

$$x + 5 = 105 \Rightarrow x = 100^\circ$$

14 Coordinate Geometry: Slope Criteria

Slope Formula

For two points (x_1, y_1) and (x_2, y_2) , the slope of the line segment connecting them is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Parallel and Perpendicular Slope Criteria

Given a line $y = mx + b$:

- A line **parallel** to it has the *same slope*: $y = mx + c$.
- A line **perpendicular** to it has the *negative reciprocal slope*: $y = -\frac{1}{m}x + c$.

Note: Two parallel lines have the same slope but different y-intercepts.

Example 4: Classify each line against $y = 17x + 8$

- | | |
|---|--|
| (i) $2y = 34x + 5 \Rightarrow y = 17x + 2.5$ | slope 17 $\Rightarrow$ parallel |
| (ii) $3y = 51x + 7 \Rightarrow y = 17x + \frac{7}{3}$ | slope 17 $\Rightarrow$ parallel |
| (iii) $-\frac{y}{7} = \frac{x}{119} + 10 \Rightarrow y = -\frac{1}{17}x - 70$ | slope $-\frac{1}{17} \Rightarrow$ perpendicular |
| (iv) $\frac{y}{5} = 17x + 9 \Rightarrow y = 85x + 45$ | slope 85 $\Rightarrow$ neither |

Example 5: Slope between two points

Find the slope of the line segment joining $(2, 5)$ and $(-5, 2)$:

$$m = \frac{2 - 5}{-5 - 2} = \frac{-3}{-7} = \frac{3}{7}$$

A line **parallel** to this segment has slope $\frac{3}{7}$; a line **perpendicular** to it has slope $-\frac{7}{3}$.

Example 6: Distance from a point to a line

Find the distance from $(2, 1)$ to the line $y = x - 1$. The given line has slope 1, so the perpendicular line through $(2, 1)$ has slope -1 :

$$y = -x + c \Rightarrow 1 = -2 + c \Rightarrow c = 3 \Rightarrow y = -x + 3.$$

Note: The source notes set up the perpendicular line through the point but stop before solving for the

intersection point and final distance. Completing it: solving $x - 1 = -x + 3$ gives $x = 2, y = 1$ – the point itself, meaning $(2, 1)$ actually lies on the line $y = x - 1$ (distance 0). Worth double-checking against the original lesson, since a distance-to-a-line example is usually built around a point that is not on the line.